

RYCROFT AGCM, AB, Canada

WMO: 715460

Lat: 55.7769N

Lon: 118.6681W

Elev: 1886

StdP: 13.72

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.3	-22.8	-35.0	0.8	-28.0	-29.0	1.2	-22.6	30.0	37.5	27.4	37.0	4.6	200	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	82.5	60.8	79.3	59.8	76.3	58.8	64.4	76.4	62.7	75.0	60.9	72.4	8.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	83.2	68.8	58.1	77.2	66.7	56.3	72.4	64.8	30.5	76.0	29.2	75.8	27.9	72.4	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	20.7	18.1	-34.9	88.0	4.3	3.3	-37.9	90.3	-40.4	92.2	-42.8	94.1	-45.9	96.5		
(5)			-35.0	67.5	4.3	3.2	-38.1	69.8	-40.6	71.7	-43.0	73.4	-46.1	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	12.7	11.7	21.6	37.9	51.3	58.4	61.5	59.8	51.2	38.5	19.8	10.0	(7)	(8)
	DBStd	22.40	16.79	16.52	14.63	9.24	7.80	5.44	4.29	5.90	7.82	9.16	14.08	15.15		
	HDD50	6154	1155	1072	880	370	79	5	0	4	77	366	905	1241		
	HDD65	10504	1620	1492	1345	812	427	208	124	178	416	821	1355	1706		
	CDD50	1170	0	0	0	8	118	257	356	308	113	10	0	0		
	CDD65	45	0	0	0	0	1	10	15	17	2	0	0	0		
	CDH74	1202	0	0	0	8	106	223	389	382	93	1	0	0		
	CDH80	234	0	0	0	1	17	40	67	95	14	0	0	0		
	WSAvg	7.4	7.5	7.0	7.0	8.6	9.0	8.2	6.8	6.5	7.3	7.9	6.8	6.9		
	PrecAvg	18.5	1.3	0.8	0.9	1.0	1.8	3.0	2.9	2.1	1.5	1.2	1.2	0.9		
(15) Precipitation	PrecMax	25.3	4.0	2.0	1.8	1.9	3.5	5.4	5.2	5.5	4.3	3.5	3.6	3.8	(16)	(17)
	PrecMin	12.1	0.4	0.2	0.4	0.2	0.7	0.7	1.1	0.7	0.3	0.2	0.2	0.0		
	PrecStd	3.6	0.7	0.4	0.4	0.5	0.8	1.4	1.1	1.1	0.8	0.7	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	46.1	52.8	72.0	81.9	84.8	86.1	87.7	81.5	70.6	50.1	43.4	(20)	(21)
		MCWB	38.4	39.7	43.1	51.4	55.6	61.3	63.9	62.6	58.5	53.2	41.9	36.7		
	2%	DB	42.6	41.3	47.3	63.6	76.7	79.4	81.2	82.5	76.5	61.1	44.5	40.0		
		MCWB	37.0	35.9	39.6	46.0	53.2	59.1	61.6	62.1	57.5	48.3	38.6	34.6		
	5%	DB	40.6	38.6	43.3	57.2	72.0	76.0	78.3	78.0	71.3	56.5	41.3	36.7		
		MCWB	35.9	34.0	36.7	43.2	51.3	57.7	61.1	60.7	54.9	45.5	36.2	32.3		
	10%	DB	37.7	35.4	40.0	52.8	68.2	72.5	75.2	74.4	66.5	52.9	38.3	33.1		
		MCWB	33.5	31.8	34.7	41.2	50.0	56.2	59.8	59.5	53.0	43.6	34.2	29.1		
	0.4%	WB	39.1	39.8	44.7	52.2	58.1	64.7	68.1	66.6	62.0	54.5	43.0	37.5	(27)	(28)
		MCDB	44.4	45.4	52.3	72.5	77.1	78.4	78.7	79.4	77.4	67.8	47.7	42.5		
	2%	WB	37.5	36.1	40.0	47.8	54.8	61.6	64.7	64.3	58.6	49.6	39.2	34.9		
		MCDB	42.1	40.5	46.6	61.8	71.8	75.1	75.9	76.4	71.1	58.7	43.8	39.5		
	5%	WB	36.0	34.3	37.1	44.1	52.8	59.5	63.0	62.5	56.4	46.7	36.4	32.4		
		MCDB	40.4	38.2	42.5	55.5	68.7	72.1	74.2	75.2	69.7	55.0	40.4	36.7		
	10%	WB	33.7	31.6	34.9	41.7	51.1	57.8	61.3	60.6	54.1	44.2	34.4	28.9		
		MCDB	37.5	35.3	39.7	51.7	65.7	69.3	72.1	71.9	65.0	51.5	38.2	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	18.6	19.3	20.5	25.6	21.8	23.3	24.3	23.5	18.4	16.3	16.1	(36)	(37)
		MCDBR	17.4	18.6	20.2	29.9	33.0	29.4	30.8	31.6	31.5	25.3	18.5	20.1		
	5% WB	MCWBR	13.5	14.2	14.0	15.7	14.8	12.6	14.3	14.2	14.7	14.4	13.8	16.6		
		MCWBR	16.9	18.0	19.9	27.6	29.5	25.3	27.2	28.3	29.4	24.4	18.0	20.3		
		MCWBR	13.4	13.8	14.1	15.3	13.6	11.9	14.2	14.5	14.8	14.8	14.2	17.0		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.251	0.307	0.325	0.347	0.354	0.355	0.354	0.312	0.287	0.245	0.228	(41)	(42)
	taud		2.244	2.344	2.247	2.376	2.386	2.385	2.402	2.396	2.517	2.541	2.405	2.226		
	Ebn at Noon		217	269	272	285	284	282	279	272	272	254	224	197		
	Edh at Noon		17	23	32	33	35	35	34	32	25	19	14	13		
(44) All-Sky Solar Radiation	RadAvg		227	527	999	1491	1777	1931	1877	1525	1044	556	249	159	(45)	(46)
	RadStd		23	41	78	82	120	153	116	135	91	63	22	13		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page